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SUPERSEDING
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21 March 1952

FEDERAL SPECIFICATION

PYRIDINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers pyridine used in denaturing alcohol and as a solvent for anhydrous salts, rubber, paint, etc.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specification and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

VV-L-791—Lubricants, Liquid Fuels, and Related Products; Methods of Inspection, Sampling, and Testing.

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by check, money order, or cash, to the General Services Administration, Business Service Center, Region 3, Room 7110, Seventh and D Streets, S. W., Washington 25, D. C. Single copies of product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle. Prices of additional copies may be obtained from the GSA Regional offices.

(Copies of the Military standard referenced above, required by the contractor in connection with specific procurement functions, should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Pyridine shall be a technical grade and shall be miscible with water in all proportions.

3.2 **Specific gravity.**—The specific gravity of pyridine at 77°F. (25°C.) shall be between 0.970 and 0.980 when tested as specified in 4.4.2.

3.3 **Nonvolatile matter.**—The proportion of the material which does not volatilize at 260°F. shall not exceed 0.05 percent when tested as specified in 4.4.3.

3.4 **Sulfate.**—The material shall contain not more than 0.006 percent of sulfate, (SO_4) when tested as specified in 4.4.4.

3.5 **Distillation range.** — Upon distillation the sample shall conform to the following limits when tested as specified in 4.4.1:

(a) When the first drop falls from the end of the condenser, the thermometer shall read not less than 225°F. When 10 percent has been recovered

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in the receiver, the thermometer shall read not less than 235°F.

(b) When 90 percent has been recovered in the receiver, the thermometer shall read not more than 250°F.

(c) The end point shall be not higher than 290°F.

3.6 Color.—When tested as specified in 4.4.1, the color shall be not darker than No. 1 of method 10.2 of Federal Specification VV–L–791.

3.7 Odor.—Pyridine shall have a strong odor characteristic of the material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Lot.*—For purposes of sampling a lot shall consist of one distillation batch.

4.1.2 *Sampling for tests (at a Government laboratory).*—From each inspection lot (see 4.1.1) the Government inspector shall take two containers at random. They shall be clearly marked and forwarded to the testing laboratory designated by the procuring activity.

4.1.3 *Sampling for inspection of containers (at the contractor's plant).*—A random sample of filled containers shall be taken by the Government inspector in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level = 2.5 percent defective to verify compliance with this specification regarding fill, closure, marking, and other requirements not involving tests.

4.2 Inspection.

4.2.1 *Inspection of containers (at the contractor's plant).*—Each sample-filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed (or repaired) all nonconforming containers.

4.3 Lot acceptance tests.—The sample specimens selected in accordance with 4.1.2 shall be subjected to the tests specified in 4.4. Lots shall be accepted or rejected by the Government inspector on the basis of the laboratory test report on the transmitted samples. Rejected lots may be resubmitted for acceptance tests, provided the contractor has removed (or reworked) all nonconforming products.

4.4 Test procedures.

4.4.1 The following tests shall be in accordance with the applicable methods of Federal Specification VV–L–791:

Test	*Method*
Color	10.2
Distillation	100.1

4.4.2 *Specific gravity.*—The specific gravity at 77°/77°F. shall be determined by weighing a sample of the material in a calibrated pycnometer.

4.4.3 *Nonvolatile matter.*—Weigh 100 grams of the material into a tared evaporating dish, evaporate to dryness, and dry at 260° ± 3°F. The weight of residue shall not exceed 0.05 gram.

4.4.4 *Sulfate.*—Dilute 10 grams of the material with 100 milliliters of distilled water, acidify with 1 milliliter of hydrochloric acid, and heat to boiling. To the boil-

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ing solution add 10 milliliters of 10 percent barium-chloride solution. Allow the sample to cool and the precipitate to settle for 16 hours. Filter, wash, ignite, and weigh the precipitate formed. The weight of the precipitate shall not exceed 1.5 milligrams.

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified, commercial packages are acceptable.

5.2 Packing.—Pyridine shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.—Each container shall be durably and legibly labeled with the following:

WARNING! FLAMMABLE.
Keep away from heat and open flame.
Keep container closed.
Use with adequate ventilation.
Avoid prolonged breathing of vapor.

In addition, shipping containers shall be marked with the name, type, and size of the article as defined by the contract or order under which the shipment is made, the quantity of each contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the title, number, and date of this specification and any desired options offered herein.

6.2 Unit of purchase.—Pyridine is purchased by weight, the unit being an avoirdupois pound.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.3 Transportation description.

6.3.1 *Transportation description applicable to these items is:*

Pyridine.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standards.*

Federal Specifications:

NN–B–591—Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).
NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
NN–B–631—Boxes; Wood, Wirebound (for Domestic Shipment).
QQ–S–781—Strapping, Flat; Steel.
VV–L–791—Lubricants, Liquid Fuels, and Related Products; Methods of Inspection, Sampling, and Testing.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes, Fiber, Solid (for Domestic Shipment).

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Military Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood-Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

MIL–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

MIL–C–124—Containers (Cans, Pails, and Drums) Metal (for Other than Subsistence Items).

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN–P–139—Packaging and Packing for Overseas Shipment — Plywood, Container Grade.

MIL–A–140—Adhesive, Water-Resistant, Waterproof Barrier Material.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Preparation for delivery.*

7.2.2.1 *Packaging.*—Pyridine shall be furnished in screw top cans containing 5 pounds each. Cans shall conform to Military Specification MIL–C–124.

7.2.2.2 *Packing.*

7.2.2.2.1 *For domestic shipment.* — Pyridine, packaged as specified in 7.2.2.1, shall be packed in wood cleated fiberboard, cleated plywood, nailed wood, or wirebound, corrugated or solid fiberboard boxes conforming to Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631 and LLL–B–636, respectively. Fiberboard boxes shall conform to the special requirements of the applicable box specification. Closure of the boxes shall be made with adhesive, metal fastenings, or tape, or a combination of these methods in such a manner that the closures of the boxes will not break nor open when the box is tested in accordance with the applicable container specification. The gross weight of wood boxes shall not exceed 150 pounds; and of fiberboard boxes, shall not exceed the weight limitations of the applicable box specification.

7.2.2.2.2 *For overseas shipment.* — Pyridine, packaged in unit containers as specified in 7.2.2.1, shall be packed in wood cleated plywood, nailed wood, wirebound or fiberboard boxes conforming to style A or B of Military Specification JAN–P–105, style 2, 2½, or 3 of Military Specification JAN–P–106, MIL–B–107, or symbol V3c or V3s of Military Specification JAN–P–108, respectively. Plywood shall conform to type A or B, condition I of Military Specification JAN–P–139. Boxes shall be lined with a sealed waterproof caseliner made of material conforming to Military Specification JAN–P–125. Liners shall be sealed with an adhesive conforming to Military Specification MIL–A–140. Seams and closures shall have strength and water resistance equal to that of the body material, and shall have a continuous seam at least ¾ inch wide. Shipping containers shall be closed and strapped in accordance with the applicable container specification, using flat steel strapping conforming to type I, class A or B, ⅝ or ¾ inch by 0.020 inch thick of Federal Specification QQ–S–781. The gross weight of wood boxes shall not exceed 150 pounds; of fiberboard boxes, 70 pounds.

7.2.2.3 *Marking.*—In addition to the labeling specified in 3.8 and any special marking required by the contract or order, interior

[1] Applicable to Navy purchases only.

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packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

7.2.3 *Inspection procedures (Navy only).* —General inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.4 *Inspection (Air Force only).* — The contractor shall furnish all samples and shall be responsible for accomplishing the inspection tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring service. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of the material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

NAVY INTEREST: Sh S Y

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

Copies of this specification can be purchased for 5 cents each from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. Single copies required for bidding purposes are available without charge at the nearest GSA Regional Office.

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